# Analog Signal Conditioning and Filter Design for Mechatronic Systems

Lab 5 –ENME 561/661



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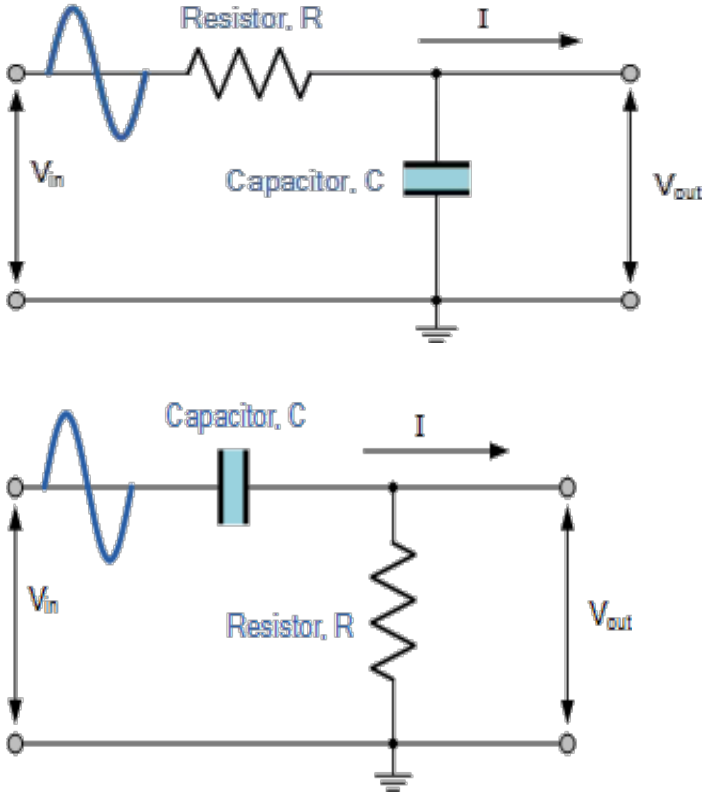
# What this lab asks you to do

- You generate one signal that contains both what you want and what you do not:  sin(10t) + 0.5 sin(100t)
- 5.1  Passive filters
  - Identify the filter type, choose a cut-off frequency from the resistors provided, build the RC circuit, and compare it against the same transfer function modelled in Simulink. Then convert it to a high-pass filter.
- 5.2  Active filters
  - Add an LM741 to your best passive filter and compare the two.
- 5.3  Active vs. passive
  - Repeat with a weak signal, 0.1 sin(10t), that the noise dominates, and use gain to recover it.
- 5.4  Op-amp circuits
  - Find the saturation limits of a unity-gain buffer, then design a circuit that computes Vout = (2V1 - V2) / 3.
- Every circuit is built on the breadboard and photographed, and every result is compared against theory.

# RC Filters

- Low-pass filter
  - To pass frequencies below the cut-off frequency
- High-pass filter
  - To pass frequencies above the cut-off frequency
- In both cases, the cut-off frequency is

$$w_c = \frac{1}{RC}$$




Images: https://www.electronics-tutorials.ws/filter/filter_3.html

# Op-Amp Circuits

- Inverting Op-Amp

$$\frac{V_{out}}{V_{in}} = -\frac{R_F}{R_{in}}$$

- Non-Inverting Op-Amp

$$\frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_2}$$

- Summing Op-Amp

$$V_{out} = -\left(V_1 * \frac{R_f}{R_1} + V_2 * \frac{R_f}{R_2} + V_3 * \frac{R_f}{R_3}\right)$$

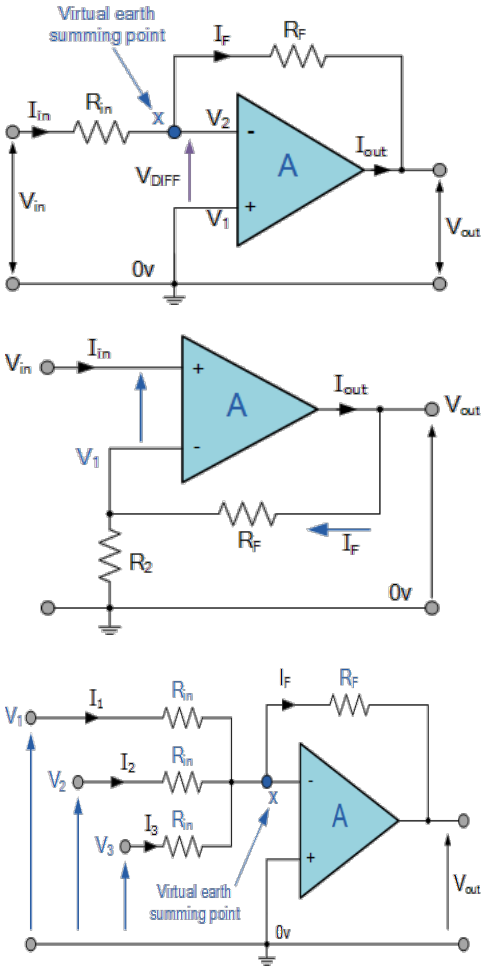
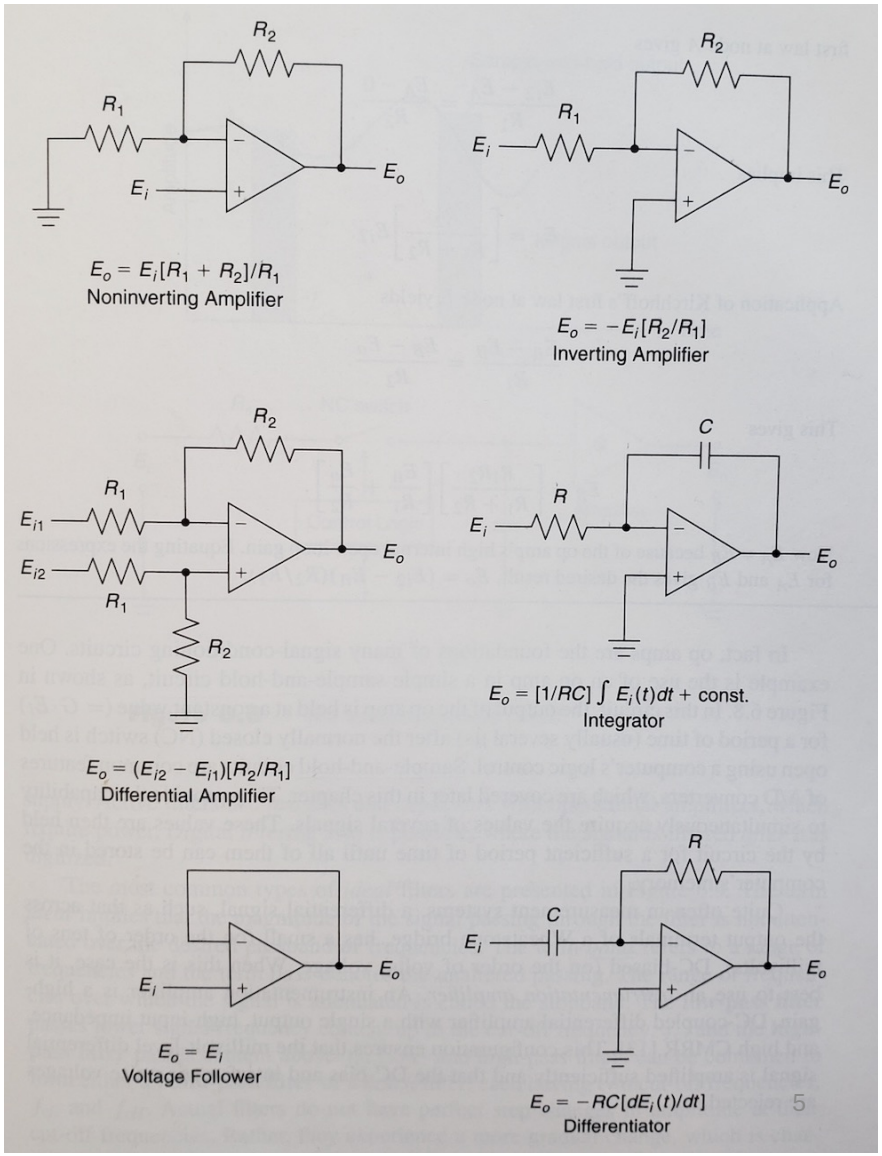


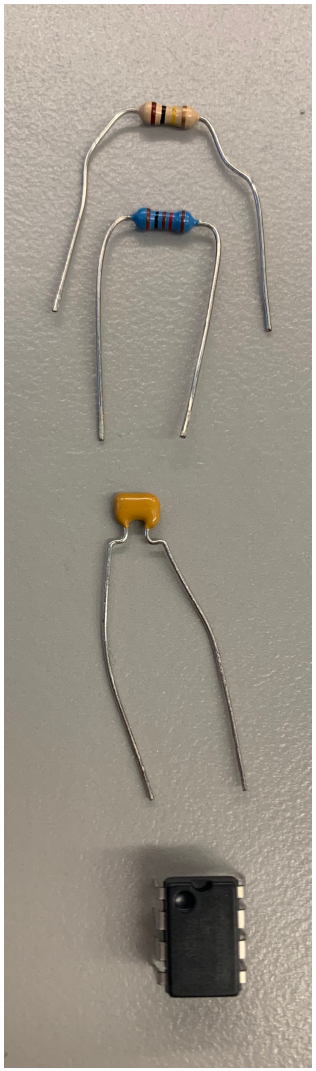

Image credit: https://www.electronics-tutorials.ws/opamp/opamp_2.html

# Popular op-amp configurations

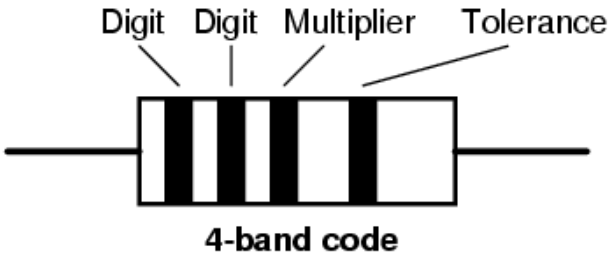


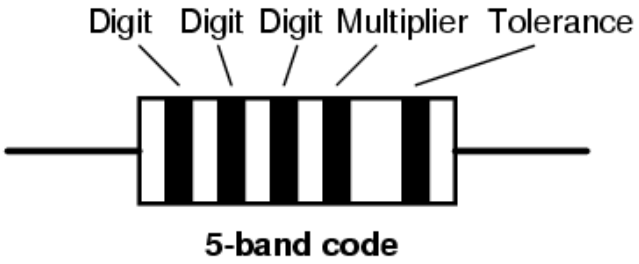
# Electrical Components

- Resistors
  - 8 different values. The value is denoted by the colored bands. *Bags are not all identical — work from the resistors in your own bag, not another group's.*
- 1 Capacitor
  - One size (0.33 $\mu$F) will be used in this lab
- 4 Op-Amps provided
  - LM741C – see data sheet for full info *(the design question in Section 5.4 allows you up to 3)*



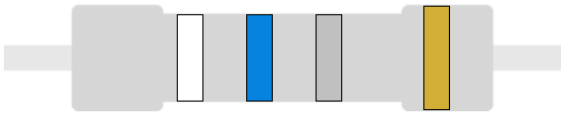
# Resistor Color Codes




Example:

4 Band Resistor



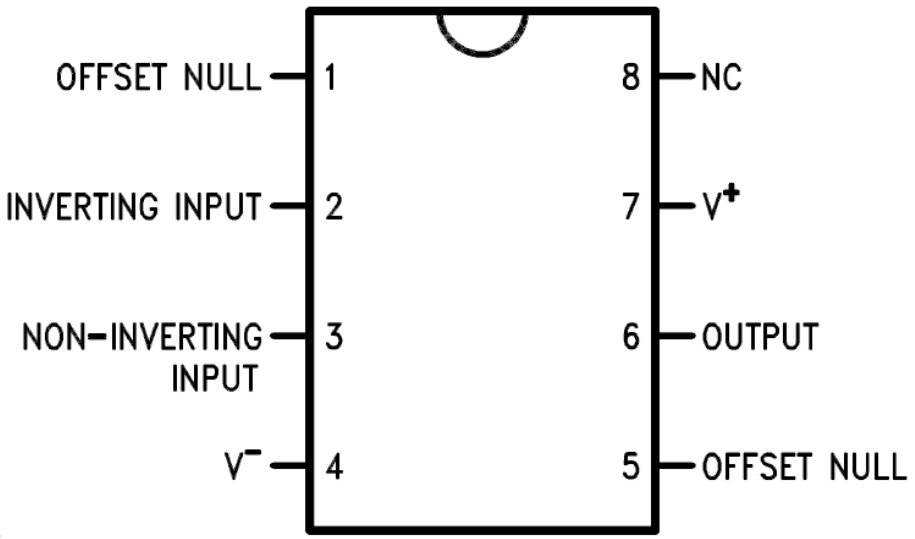
| 1st Digit | 2nd Digit | Multiplier | Tolerance |
|---|---|---|---|
| 9 White | 6 Blue | ÷100 Silver | ± 5% Gold |

| | | |
|---|---|---|
| Resistance: | 0.96 | ohms |
| Tolerance: | ± 5% | |
| Minimum: | 0.912000 | ohms |
| Maximum: | 1.008000 | ohms |

Resistor Color Code Calculator: https://www.allaboutcircuits.com/textbook/reference/chpt-2/resistor-color-codes/

Image: https://www.allaboutcircuits.com/tools/resistor-color-code-calculator/

# LM741CN Op Amp



# Breadboard

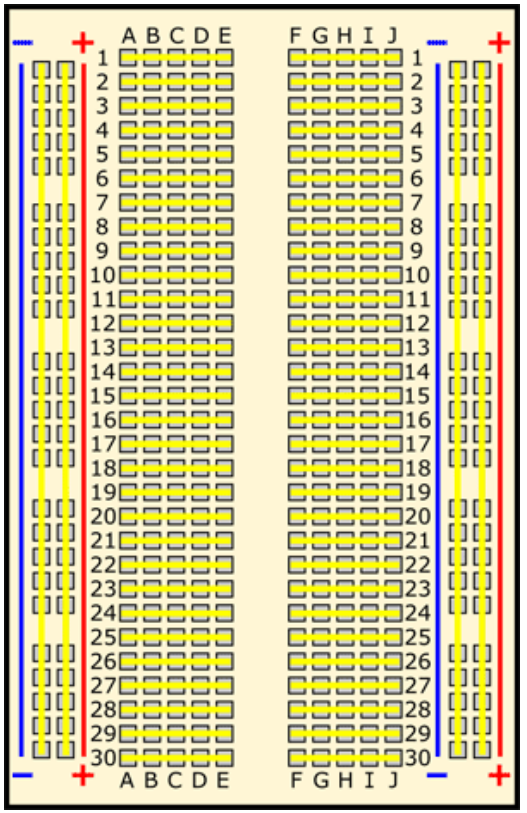
To **prototype** an electronic circuit



Image Credit: https://www.sciencebuddies.org/science-fair-projects/references/how-to-use-a-breadboard

## How are the holes connected?

Highlighted in yellow lines



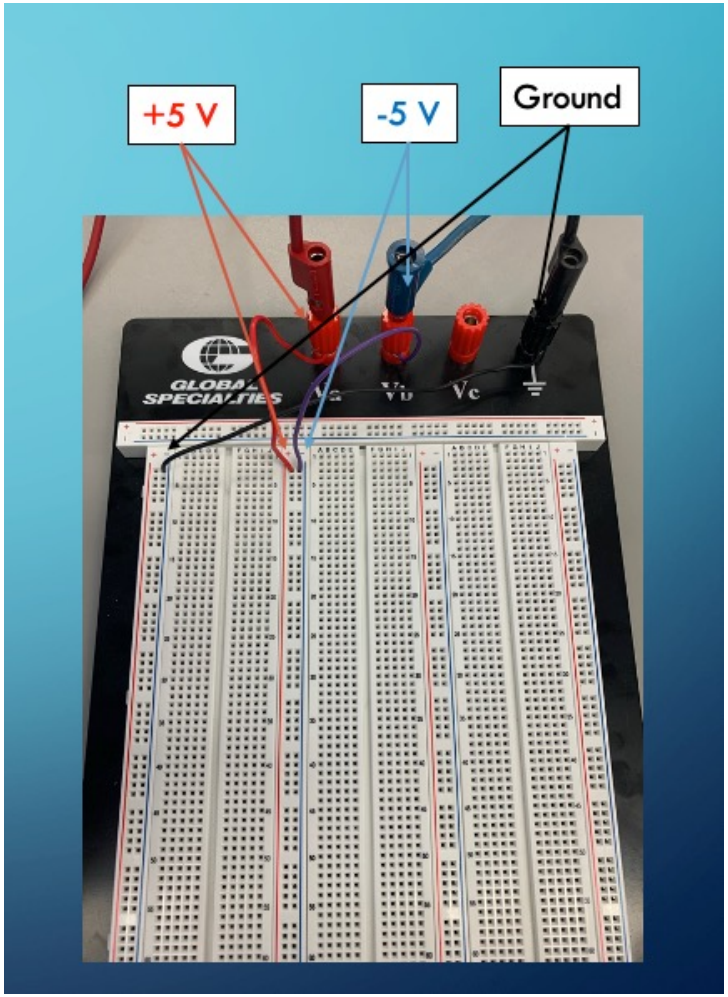

Buses to supply power:

Blue represents the Ground or negative bus

Red represents Positive bus

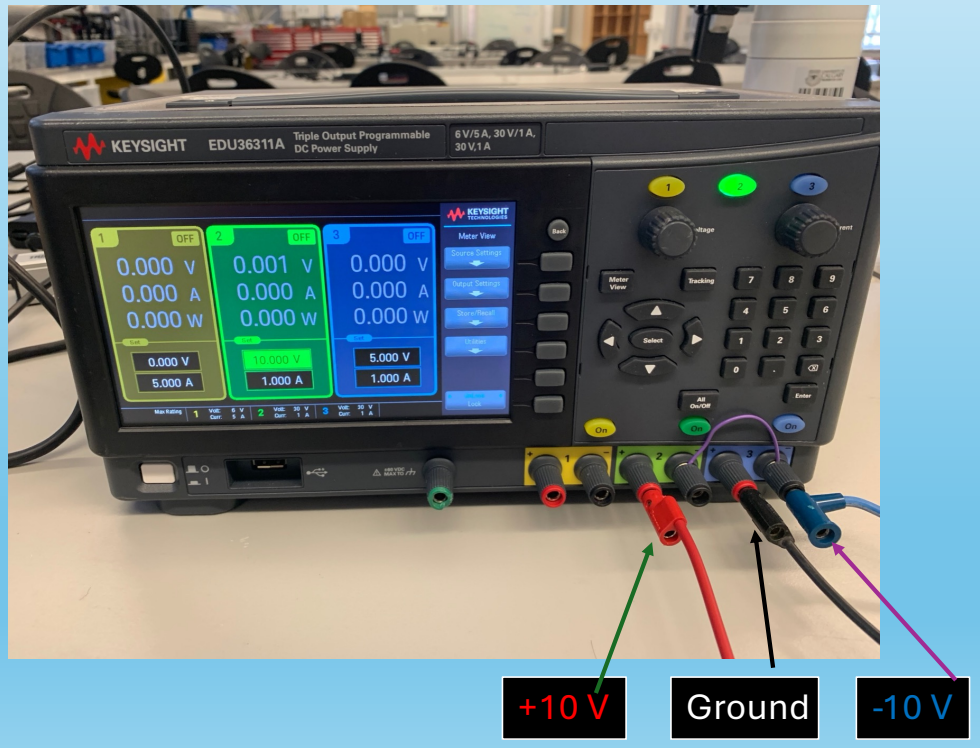
# Externally Powered Breadboard

- Should be wired to the power supply as shown right and power should be connected in the columns as shown
- Make sure that the filter circuit, DAQ and Power supply have a common ground.



# Power Supply

- To provide the power to build active circuits using op-amp
- Channel 2 needs to be set to output 10 V at 1 A
- Channel 3 needs to be set to output -10 V at 1 A
- Connect GND properly
- This will effectively result in a range of -10 V to 10 V for the power supply to the op-amp



# Power supply




- *1 display*
- *2 Output selection keys*
- *3 Voltage/Current knobs*
- *4 Meter View and Tracking keys*
  - *– [Meter View] turns on the meter view for the selected output.*
  - *– [Tracking] turns on or off the track mode for output 2 and 3.*
- *5 Navigation keys. – Navigate through the control dialog windows; press [Select] key to select a control.*
- *6 Numeric keypad*
  - *–Enters numeric values. Press [Enter] key to complete the entry.*
  - *–Deletes the values entered into the dialog using the back key.*
- *7 All Output On/Off key*
- *Turns all outputs On and Off according to the specified turn-on and turn-off delays.*
- *8 Output On key- Turns individual outputs On or Off; outputs are on when the key is lit.*
- *9 Output terminals*
- *10 Softkeys*
- *Accesses the soft front panel menu. The Back key moves back in the softkey menu hierarchy.*
- *11 Earth ground reference*
- *12 USB port*
- 13 Power switch

*Source: Keysight EDU36311A User's Guide*

# Controlling the Outputs

***Step 1*** *- Select an output*




- *Press one of the output selection keys to select an output to control. The lit key identifies the selected output. All subsequent output-specific front panel commands are sent to the selected output.*

***Step 2*** *- Set the output voltage and current*




- *Turn the Voltage and Current knobs. The output voltage or current setting changes when they are turned.*
- *You can also enter the voltage and current values directly in the numeric entry fields (the Set fields) in the meter view display. There are three methods to enter the value:*
  - *– Numeric keys: Use the navigation keys to select the field; use the numeric entry keys to enter the value. The value becomes active when you press [Enter].*
  - *– Navigation keys: Use the navigation keys to select the field, press [Select] to edit. Use the left or right arrow to select the area of value edit. Use the up or down arrow to increase or decrease the value. The value becomes active when you press [Select] or [Enter].*
  - *– Voltage and Current knobs: use the Voltage and Current knobs to adjust the values in the Voltage and Current fields.*

***Step 3*** *- Enable the output*

- *Press the color-coded [On] key to enable an individual output. When an output is on, the [On] key for that output is lit. When an output is off, the [On] key is not lit. The [All On/Off] key turn all outputs on or off simultaneously.*
- *The state of a disabled output (output off) is a condition of zero output voltage and zero source current.*

***Step 4*** *- View the output voltage and current*

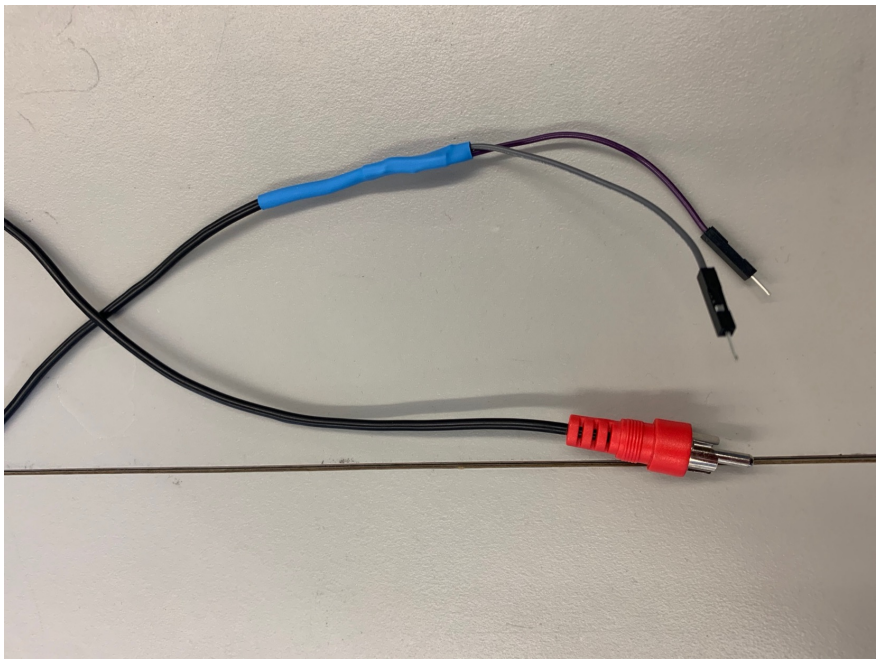
- *Press [Meter View] to view the output voltage and current. When an output is enabled, the front panel meters continuously measure and display the output voltage and current.*



*Source: Keysight EDU36311A User's Guide-page 43*

# RCA to Pin Cables

- Custom Cables for connecting from the DAQ to the breadboard
- RCA plugs into either analogue output or input and the pin goes to the breadboard
- A convention is the darker pin is the ground and the lighter is the signal
  - We use Purple, Black, and Blue for ground

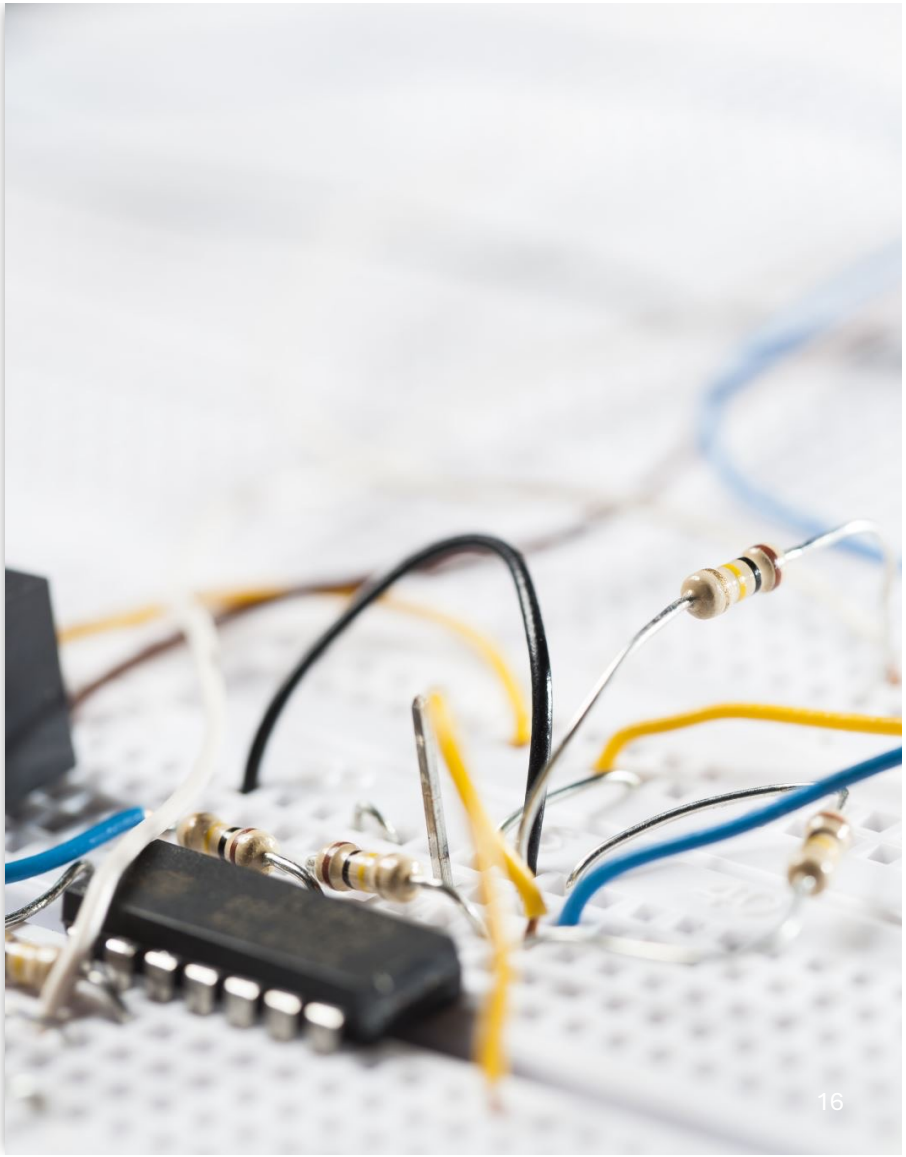


# Supply Cart

- For the later section of this lab, you will be designing custom circuits and will have to collect the supplies yourself. All supplies will be found on the cart
- If you need wire to make connections, there are jumper wires, spools and wire stripers on the cart
  - There are also small plastic bags, for you to keep the wire you cut in between lab sessions
- All other components you need to make the circuit will be in plastic bags on the cart
  - Make sure to put the components back in the same bag you got them from
- If you think a component, you are using has been damaged let either a TA or lab tech know

# Electrical Equipment Safety Notes

- Make sure the power supply is off when installing or manipulating components on the breadboard
- Do not apply higher voltages than specified in the lab manual/this presentation
- Make sure that the filter circuit, DAQ and Power supply have a common ground.
- If you are unsure of your circuit design, have a TA or a Lab Tech check it before you apply the power
- Use only the power cord that was supplied with your instrument. Using other types of power cords may cause overheating of the power cord, resulting in fire.
- Turn off AC power before connecting the wires to the front panel. All wires and straps must be properly connected with the binding posts securely tightened.
- **In this lab the only supply settings you should ever use are +10 V on channel 2 and -10 V on channel 3, both at 1 A. Do not raise either channel above 10 V.**

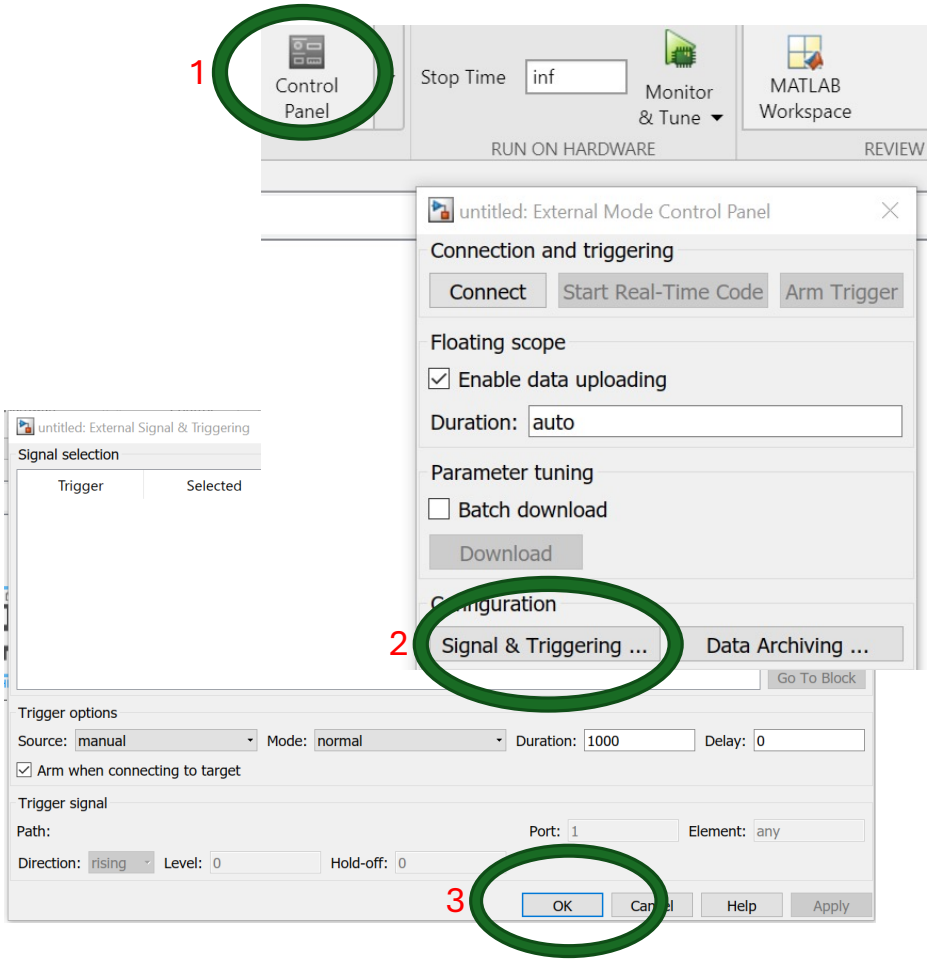


# Required Simulink configuration

- Set all of this before you build anything. It is the same for every exercise in this lab.
- HIL Initialize → board type: q8_usb (not qube_servo2_usb)
- Configuration Parameters → Solver → Type: Fixed-step, Solver: ode1 (Euler)
- Fixed step size: 0.001 s
- HIL Write and HIL Read → Sample Time: 0.001 s
- Wiring: AO #0 → your circuit input, your circuit output → AI #0. The darker lead (black / purple / blue) is ground.
- **Why 1 kHz: the fastest thing in this lab is 100 rad/s ≈ 15.9 Hz, so a 1 kHz sample rate is about 31 times the Nyquist rate. Anything you see in the data is the filter, not aliasing.**
- **Check these before your first run. A wrong solver does not throw an error — it silently changes every result you record.**

# Simulink to MATLAB

- **To Workspace** block can be used, need to use save format of Array not Timeseries
- You will likely also have to increase the amount of data points (10000 will be enough for this lab)



# Tools to draw circuit diagrams

Use services like https://www.circuit-diagram.org

Hand-drawn circuits are not acceptable



# Plotting Single-Sided Amplitude Spectrum of a Signal

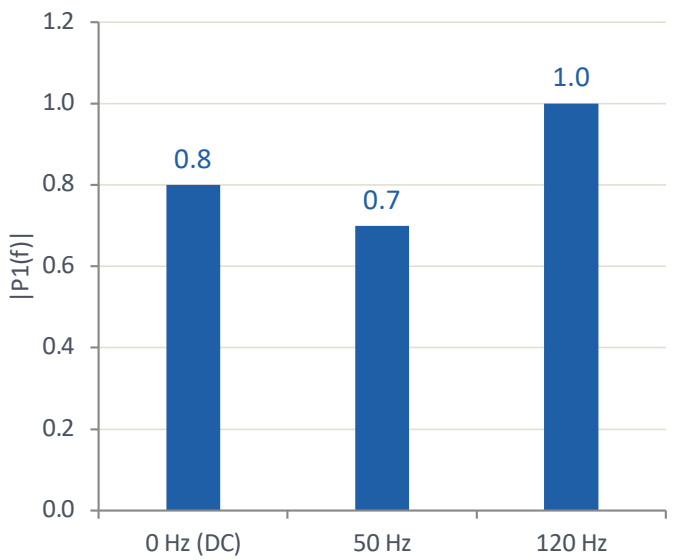
Example: Fs = 1 kHz, L = 1500;
S = 0.8 + 0.7 sin(2π·50t) + sin(2π·120t)

1. `P2 = abs(Y/L);`
   Divide by L (fft is unscaled), take magnitude

2. `P1 = P2(1:L/2+1);`
   Keep 0 to Fs/2: the positive frequencies

3. `P1(2:end-1) = 2*P1(2:end-1);`
   Double, except DC and Nyquist (no mirror twin)

4. `f = Fs/L*(0:(L/2)); plot(f,P1)`
   Bins sit Fs/L apart, from 0 up to Fs/2




Source: MathWorks, fft documentation:
https://www.mathworks.com/help/matlab/ref/fft.html

# Tools to design filters- optional

**Exam note: you must be able to derive the transfer function of every filter covered in lecture, in the lab, and in the D2L reading materials.**

- Filter Designer: Web-based design tool from Texas Instruments

https://www.ti.com/tool/download/ANALOG-FILTER-DESIGNER/1.0

- Analog filter design by Analog Devices, Inc.

https://tools.analog.com/en/filterwizard/

- FilterLab Active Filter Designer

https://www.microchip.com/en-us/tools-resources/develop/analog-development-tool-ecosystem/filterlab